CHECK WHAT YOU LEARNT

Check Yourself - Unit 5

A. Choose the correct answer

1. An H-bridge lets a circuit
   (a) measure distance (b) reverse a motor (c) store power (d) read a sensor
2. On the L298N, ENA and ENB control
   (a) direction (b) speed (c) the sensor (d) the battery
3. The wire most often forgotten is
   (a) TRIG to D11 (b) Arduino GND to L298N GND (c) the LED wire (d) the USB cable
4. If the motors ignore your speed setting, you should
   (a) change the code (b) remove the ENA and ENB jumper caps (c) use a bigger battery (d) swap the motors
5. pulseIn(echo, HIGH, 25000) will give up after
   (a) 25 microseconds (b) 25 milliseconds (c) 25 seconds (d) never
6. A reading of 0 from the sensor is converted to 400 because
   (a) 400 is the maximum (b) no echo means nothing is nearby (c) it makes the robot faster (d) the code needs it

B. Fill in the blanks

1 Two pins set a motor's ______________ and one pin sets its ______________.

2 IN1 HIGH and IN2 LOW makes the motor turn ______________.

3 A wheel spinning the wrong way is fixed by swapping its two ______________.

4 An ordered set of steps a machine follows is called an ______________.

C. Answer in one or two lines

1 Explain in your own words why the Arduino and the L298N must share a ground.

2 Your robot curves to the left while driving forward. Give two possible reasons.

3 Write down the turn delay you settled on and how you arrived at it.

4 Which of the three projects this year taught you the most, and why?

Extra Challenges

Finished early? Pick any of these. They all use parts you already have.

#	Challenge	Uses
1	A reversing alarm that beeps faster as an object comes closer	HC-SR04 + buzzer
2	A height station that reports feet and inches as well as centimetres	maths
3	A digital tape measure that holds the reading when you press a button	switch + LEDs
4	A servo that follows your hand: closer hand, larger angle	HC-SR04 + servo
5	A bin that also reports how full it is	second HC-SR04
6	A door that opens for people but stays shut for walls	servo + threshold
7	A robot that drives a square path with no sensor at all	L298N + timing
8	A robot that chooses left or right by sweeping a servo-mounted sensor	servo + HC-SR04
9	A speed test: measure how long the robot takes to cross 2 metres	millis()
10	A robot that beeps and flashes before every turn, like a real vehicle	buzzer + LEDs

MIXED REVISION

Twenty Questions to Test Yourself

1. What are the three steps every robot follows?
2. Why must a circuit be a complete loop?
3. How many holes are joined in one breadboard column?
4. How far does sound travel in one microsecond?
5. Write the four steps of the ultrasonic trigger sequence.
6. What does pulseIn() actually measure?
7. Why does the distance formula divide by two?
8. An echo returns in 1500 microseconds. How far away is the object?
9. Why should a timing value be stored in a long rather than an int?
10. What is a derived quantity? Give one example from this book.
11. What is noise, and how does averaging reduce it?
12. How does a servo know it has reached the angle you asked for?
13. Why must you never force a servo horn by hand?
14. Give two causes of a false trigger on the smart dustbin.
15. Why can a motor not be connected directly to an Arduino pin?
16. What does an H-bridge do?
17. Which L298N pins set direction, and which set speed?
18. Why must the Arduino and the L298N share a ground?
19. Why is a 0 reading from the sensor converted to 400 in the robot code?
20. Name one improvement you would make to your robot with one more session.

Words to Know

Word	Meaning
Actuator	Any part that moves or acts: a motor, a buzzer, a lamp.
Algorithm	A clear, ordered set of steps a machine follows to solve a problem.
Analog	A value that can be anything within a range, not just on or off.
Averaging	Taking several readings and using their mean to reduce noise.
Breadboard	A block of holes for joining parts without soldering.
Current	How much electricity flows each second. Measured in amps.
Debugging	Hunting down the reason something does not work.
Derived quantity	A useful number worked out from something else that was measured.
Digital	A signal with only two values: HIGH or LOW.
Error	The difference between a measured value and the true value.
False trigger	The machine acting when nobody intended it to.
Function	An instruction you write yourself and then reuse.
Ground (GND)	The return path. Every circuit needs one.
H-bridge	Four switches that let a circuit reverse a motor on command.
Horn	The plastic arm attached to a servo's shaft.
long	A variable type that holds very large whole numbers.
Microsecond	One millionth of a second.
Motor driver	A board that switches heavy motor current on a small signal's command.
Noise	Small random variation in a sensor's readings.
pulseIn()	Measures how long a pin stays HIGH, in microseconds.
PWM	Fast switching that makes an output appear to be in between on and off.
return	Hands a value back from a function to whoever called it.
Servo	A motor that turns to an exact angle and holds it.
Shared ground	The wire that lets two power supplies agree on what zero volts means.
Threshold	The number at which a decision changes from one answer to the other.
Time of flight	Measuring distance by timing how long a signal takes to return.
Timeout	Giving up on waiting so a program is never stuck.
Ultrasonic	Sound too high for human ears to hear.

Answer Key

Section A only. Sections B and C are for you and your trainer to discuss - there is often more than one good answer.

Check Yourself - Unit 1

Q	Answer	The correct option
1	(c)	a complete loop
2	(c)	the plus side
3	(a)	columns of five
4	(b)	analog
5	(a)	20 milliamps
6	(b)	error

Check Yourself - Unit 2

Q	Answer	The correct option
1	(b)	a millionth of a second
2	(b)	how long the pin stays HIGH
3	(c)	long
4	(b)	return
5	(b)	it draws far too much current
6	(b)	5 times

Check Yourself - Unit 3

Q	Answer	The correct option
1	(b)	timing an echo
2	(b)	sound travels there and back
3	(b)	10 microseconds
4	(b)	TRIG and ECHO swapped
5	(b)	138 cm
6	(b)	noise

Check Yourself - Unit 4

Q	Answer	The correct option
1	(b)	knows its own angle
2	(c)	orange
3	(b)	turn to 90 degrees
4	(b)	pulseIn returns 0 when no echo comes back
5	(b)	the lid is too heavy
6	(a)	false trigger

Check Yourself - Unit 5

Q	Answer	The correct option
1	(b)	reverse a motor
2	(b)	speed
3	(b)	Arduino GND to L298N GND
4	(b)	remove the ENA and ENB jumper caps
5	(b)	25 milliseconds
6	(b)	no echo means nothing is nearby

My Project Log

Fill one row after every class. This is your evidence, and it counts towards your assessment.

Session	What I built or learnt	A problem I hit	How I fixed it
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

A last word

This year your machines learnt to measure honestly and then to move. Notice how much of the work was not writing code at all: mounting a sensor straight, keeping a lid light, remembering one ground wire. That is real engineering. The code is usually the easy part.